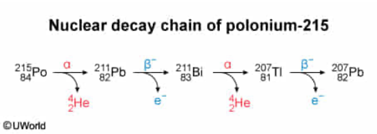
At the end of the actinium nuclear decay series, polonium-215 decays in an alternating sequence of alpha and beta decays. What are the nuclei produced in the nuclear decay sequence listed in order of formation from first to last?

- A. $^{215}_{84}\mathrm{Po} \xrightarrow{\alpha} {}^{211}_{82}\mathrm{Pb} \xrightarrow{\beta^-} {}^{211}_{81}\mathrm{Tl} \xrightarrow{\alpha} {}^{207}_{79}\mathrm{Au} \xrightarrow{\beta^-} {}^{207}_{78}\mathrm{Pt}$
  [16%]
- B. $^{215}_{84}\mathrm{Po} \xrightarrow{\alpha} {}^{211}_{82}\mathrm{Pb} \xrightarrow{\beta^-} {}^{211}_{83}\mathrm{Bi} \xrightarrow{\alpha} {}^{207}_{81}\mathrm{Tl} \xrightarrow{\beta^-} {}^{207}_{82}\mathrm{Pb}$
  [78%]
- C. $^{215}_{84}\mathrm{Po} \xrightarrow{\alpha} {}^{211}_{86}\mathrm{Rn} \xrightarrow{\beta^-} {}^{211}_{87}\mathrm{Fr} \xrightarrow{\alpha} {}^{207}_{89}\mathrm{Ac} \xrightarrow{\beta^-} {}^{207}_{91}\mathrm{Pa}$
  [3%]
- D. $^{215}_{88}\mathrm{Po} \xrightarrow{\alpha} {}^{211}_{86}\mathrm{Rn} \xrightarrow{\beta^-} {}^{211}_{85}\mathrm{At} \xrightarrow{\alpha} {}^{207}_{83}\mathrm{Bi} \xrightarrow{\beta^-} {}^{207}_{82}\mathrm{Pb}$
  [2%]

Correct
78% answered correctly

Explanation:



Radioactivity is the spontaneous emission of radiation (photons or particles) from an unstable atomic nucleus. Although several types of nuclear radiation exist, the three most common are alpha decay, beta decay, and gamma emission.

During alpha decay, an unstable nucleus ejects an alpha (α) particle consisting of 2 protons and 2 neutrons (a helium-4 nucleus without its electrons). The nucleus formed following alpha decay has a mass number that is **4 units less** and an atomic number that is **2 units less** than the nucleus that underwent alpha decay.

Beta decay can occur in three forms: β^- decay (electron emission), β^+ decay (positron emission), and electron capture. In all three forms, the **mass number is unchanged**, whereas the **atomic number** either **increases (β^- decay)** or **decreases (β^+ decay)** by **1 unit**.

Consequently, alpha decay of an atom of polonium-215 would form lead-211 by the following nuclear reaction:

$$^{215}_{84}\mathrm{Po} \rightarrow {}^{211}_{82}\mathrm{Pb} + {}^{4}_{2}\mathrm{He}$$

Subsequent β^- decay of the ^{211}Pb nucleus would eject a nuclear electron (e^-) and an antineutrino ($\overline{\nu}_e$), yielding a bismuth nucleus with the same mass number:

$$^{211}_{82}\mathrm{Pb} \rightarrow {}^{211}_{83}\mathrm{Bi} + e^- +$$

Alpha decay of the ^{211}Bi nucleus would, in turn, produce a thallium-207 nucleus:

$$^{211}_{83}\mathrm{Bi} \rightarrow {}^{207}_{81}\mathrm{Tl} + {}^{4}_{2}\mathrm{He}$$

Finally, β^- decay of the ^{207}Tl nucleus would give a stable lead-207 nucleus:

$$^{207}_{81}\mathrm{Tl} \rightarrow {}^{207}_{82}\mathrm{Pb} + e^- +$$

Therefore, the nuclei produced in the nuclear decay sequence listed in order of formation is:

$$^{215}_{84}\mathrm{Po} \xrightarrow{\alpha} {}^{211}_{82}\mathrm{Pb}$$

(Choices A and D) These sequences show a decrease in the atomic number after each β^- decay, but in a β^- decay, the atomic number *increases* 1 unit. Also, the atomic number of polonium is 84 rather than 88.

(Choice C) This sequence shows an increase in the atomic number following each alpha decay. However, in alpha decay, the atomic number should *decrease* by 2 units and the mass number should decrease by 4 units.

Educational objective:
In alpha decay, the mass number of an atomic nucleus decreases by 4 units and the atomic number decreases by 2 units. In all forms of beta decay, the mass number of an atomic nucleus remains unchanged, but the atomic number increases (β^- decay) or decreases (β^+ decay) by 1 unit.

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